Path	What Opens	Requires
ISO Standardization	MC-4 fourth monitoring category	CoDa + MC-4 + ISO/TC 69
Governance Diagnostics	Per-country compositional health	HUF + EITT + domain data
Fusion Self-Bootstrap	Stagnation cost visible, redirect funds transition	Gov. diagnostics + energy data
Wetlands as Necessity	Conservation = economic infrastructure	Gov. diagnostics + wetland data
Cross-Domain Transfer	40 domains, 208 exchanges, one instrument	MC-4 compendium + HUF
Error Detection	E-01 to E-17 detectable and classifiable	EITT catalogue + MC-4
AI Governance	D19 Accord: derivation traces + verification	HUF spec v2.0 + D20
Academic Integration	CoDa = required competency	ISO standard + adoption

Status	Quantitative Anchor	Next Action
Positioned	ISO/TC 69 pathway mapped, positioning doc complete	Route to TC 69 contacts
Complete	201 countries, 7 development stages, K1 distances	Extend to monthly resolution
Complete	$707B/yr stagnation, 141:1 ratio, zero by 2065	Present bootstrap model to energy policy
Complete	$7.98-39T services on $6M budget, 1:8M ratio	Route toolkit to Ramsar Secretariat
Defined	10 Tier 1, 10 Tier 2, 20 Stubs, 16 tools	Pilot first Tier 2 domain
Catalogued	17 errors, RNA velocity exemplar documented	Extend to all 40 domains
Defined	4 status levels, 4 transitions, SHA-256 anchored	Test with next AI-generated tool
Contingent	6 open EITT problems as research agenda	Depends on ISO pathway progress

Deliverable	File
HUF Spec v2.0	Chatgpt/higgins-unity-framework/huf_spec_v2.0.json
EITT Decomposition Framework	HUF-Decomposition-Dev/EITT_DECOMPOSITION_FRAMEWORK.md
HUF Construction	HUF-Decomposition-Dev/HUF_CONSTRUCTION.md
Document Manifest	HUF-Decomposition-Dev/HUF_DOC_MANIFEST.json
Collective Review Ledger	HUF-Decomposition-Dev/COLLECTIVE_REVIEW_LEDGER.md
ISO Positioning Document	MC4_ISO_Positioning_Document.docx
CoDaWork Packet v3	HUF_MC4_CoDaWork_Packet_v3.docx
Governance Ranking (201 countries)	HUF_Governance_Ranking_201_Countries.xlsx
Fusion Bootstrap Model	Sheet 9 of Governance Ranking
Wetlands Hidden Variables (doc)	The_Hidden_Variables_Wetlands.docx
Wetlands Hidden Variables (xlsx)	HUF_Wetlands_Hidden_Variables.xlsx
Ramsar Transformation Toolkit	Ramsar_Transformation_Toolkit.xlsx
The Lineage	THE_LINEAGE.md
Origin Bridge (DADC to MC-4)	ORIGIN_BRIDGE_DADC_TO_MC4.md
Unified Proposal	Unified_Proposal_CoDa_HUF.docx
Unified Briefing	HUF_UNIFIED_BRIEFING_S017.json

Status	Owner	Session
Complete	ChatGPT	S015
Reviewed	Peter/Claude	S016
Reviewed	Peter/Claude	S016
v1.0 - needs update	All	S016
Complete	Claude	S016
Complete	Peter/Claude	S016
Complete	Peter/Claude	S016
Complete (10 sheets)	Claude	S017
Complete (561 formulas)	Claude	S017
Complete	Claude	S017
Complete (6 sheets)	Claude	S017
Complete (6 sheets)	Claude	S017
Complete	Peter/Claude	S016
Complete	Peter/Claude	S016
NEW	Claude	S017
NEW	Claude	S017

Error ID	Category	Description
E-01	Wrong Geometry	Treating simplex as flat Euclidean
E-02	Closure Bias	Pearson (1897) spurious correlations in constrained data
E-03	Zero Replacement	Treating carrier absence as missing data
E-04	Partition Bias	Arbitrary SBP instead of system-justified
E-05	GM Flattening	CLR: trace carrier same weight as dominant
---	---	--- EXEMPLAR CASE ---
RNA-V	Wrong Surface	Hodge decomposition on UMAP embedding (Euclidean) instead of simplex (Fisher-Rao)

NOTE *Observations only. No judgment is made about the researchers. The g*

Observed Impact	MC-4 Remedy
Spurious drift signals near boundary	Require simplex-compatible metric (Aitchison/Fisher-Rao)
False correlations from sum constraint	Log-ratio transforms mandatory for correlation analysis
Phantom presence masks structural events	Zero = event, not missing data. Event-first protocol.
Meaningless balance coordinates	System-justified partition testing required
0.1% carrier weighted same as 60%	Weighted or selective CLR with carrier significance test
Results may contain embedding artifacts indistinguishable from signal	MC-4 standard: simplex-compatible metric or documented exception

gap in standards infrastructure created conditions for the error. Filling the gap

Metric	Value
--- FUSION GOVERNANCE ---	
Global energy stagnation cost	$707.3 billion/year
Current fusion R&D spending	~$5 billion/year
Stagnation-to-fusion ratio	141:1
Bootstrap redirect rate	5% of stagnation cost
Stagnation zero year	2065
Cumulative fusion deployed	329 GW
Learning curve	$20B to $3.8B per GW
--- WETLANDS CONSERVATION ---	
Global wetland ecosystem services	$7.98-39 trillion/year
Ramsar Convention budget	$6 million/year
Protection-to-value ratio	1:8,000,000
Peatland carbon storage	550 Gt
Flood buffering value	$8,000/ha/yr
Species on wetland 6% of land	40%
Ramsar revenue potential	$159M-$705M/yr
--- CROSS-DOMAIN ---	
Countries analysed	201
Domains in compendium	40
Portable tools	16
Certified exchanges	208
Error sources catalogued	17
Empirical proofs	3 ($p < 0.05$, 0.005, 0.001)

Source	Path Connection
HUF K1 analysis, SCC $50/tCO2	Fusion Self-Bootstrap
IAEA / national budgets	Fusion Self-Bootstrap
707.3 / 5.0	Fusion Self-Bootstrap
Model assumption (editable)	Fusion Self-Bootstrap
Bootstrap model projection	Fusion Self-Bootstrap
Bootstrap model at 5% redirect	Fusion Self-Bootstrap
10% cost reduction per doubling	Fusion Self-Bootstrap
Costanza, de Groot, TEEB	Wetlands as Necessity
Ramsar Secretariat	Wetlands as Necessity
6M / 7.98T (low estimate)	Wetlands as Necessity
Global peatland assessment	Wetlands (Hidden Variable 1)
EU flood directive data	Wetlands (Hidden Variable 3)
Biodiversity nursery function	Wetlands (Hidden Variable 4)
Data-as-currency model	Wetlands as Necessity
OWID/EMBER energy data	Governance Diagnostics
HUF spec v2.0	Cross-Domain Transfer
HUF spec v2.0	Cross-Domain Transfer
16 tools x 40 domains (minus incompatible)	Cross-Domain Transfer
EITT error catalogue	Error Detection
Fermentation, Ecology, Software	ISO Standardization